

- 5 (a) Using energy considerations, distinguish between electromotive force (e.m.f.) and potential difference (p.d.).

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- (b) Fig. 5.1 shows a temperature probe which is used to detect changes in temperature in the range of $5.0\text{ }^{\circ}\text{C}$ to $35.0\text{ }^{\circ}\text{C}$. A battery of e.m.f. 9.0 V and negligible internal resistance is connected in series with a thermistor and a $5.0\text{ k}\Omega$ resistor. The thermistor is represented by the dashed box. An ideal voltmeter measures the p.d. across the resistor. Fig. 5.2 shows the characteristic curve of the thermistor.

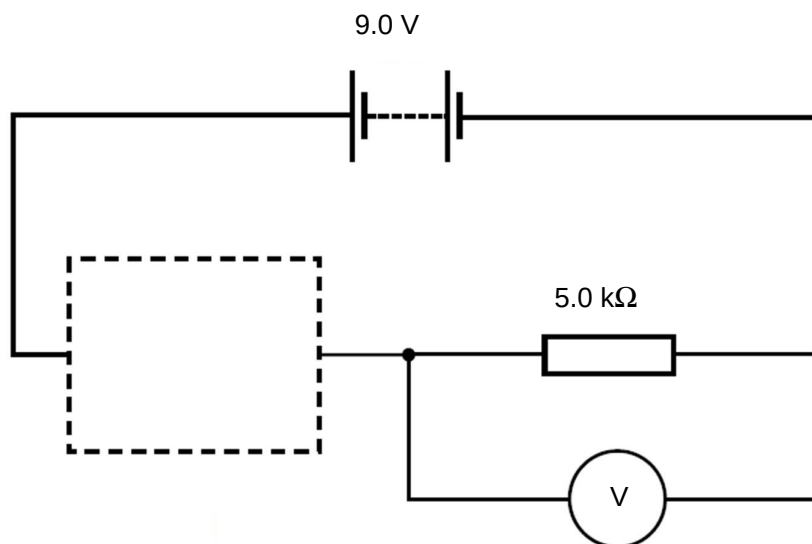
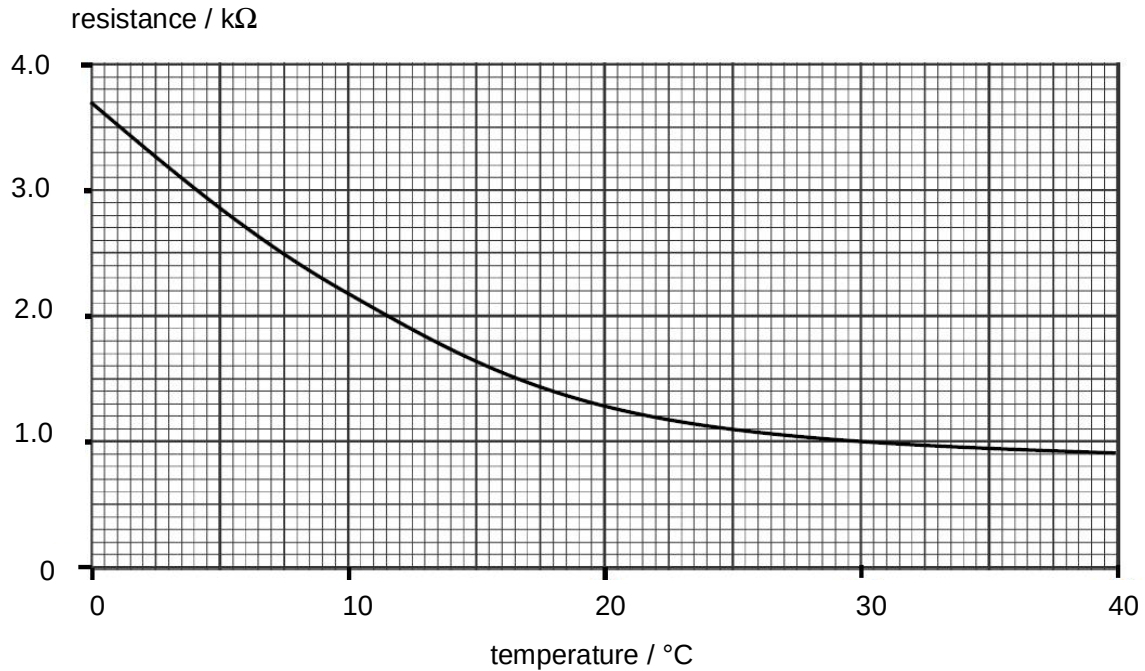


Fig. 5.1

**Fig. 5.2**

- (i) In Fig. 5.1, complete the circuit by drawing the symbol for the thermistor in the dashed box. [1]
- (ii) Using Fig. 5.2, show that the reading on the voltmeter is 5.7 V when the temperature is 5.0 °C.

[2]

- (iii) The voltmeter has a range of 0 V to 20 V and a precision of 0.1 V. When the temperature of the thermistor is 35.0 °C, the voltmeter reads 7.6 V.

State and explain a disadvantage of using this temperature probe to detect changes in temperatures in the range of 5.0 °C to 35.0 °C.

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